

-: Biomolecules :-

Introduction :-

A polyhydroxy compound that has an aldehyde or a ketone functional group, present either in free state or as a hemiacetal or hemiketal are called carbohydrate. Carbohydrate are substances having general formula $C_x(H_2O)_y$.

Classification of Carbohydrates :-

1. Monosaccharides :-

- A carbohydrate that cannot be hydrolysed to simple compounds is called monosaccharide.
- Monosaccharide which have six carbon are either aldohexoses or ketohexoses Eg Glucose, Fructose, Galactose etc.

2. Oligosaccharides :-

- Carbohydrates that yield 2 to 10 monosaccharide units, on hydrolysis, are called oligosaccharides.
- They are further classified as disaccharides, trisaccharides, tetrasaccharides etc depending upon the no. of monosaccharides, they provide on hydrolysis.
- The 2 monosaccharides units obtained by hydrolysis of a disaccharide may be same or different.
Eg Sucrose, Maltose, Lactose etc.

3. Polysaccharides :-

- A carbohydrate that can be hydrolyzed to many monosaccharide molecules is called a polysaccharides Eg Starch, Cellulose etc.
- Aldoses: Monosaccharides containing aldehyde group are called aldoses.
Ketoses: Monosaccharides containing ketonic group are called ketoses.

Sugars & Non-Sugars :-

- Both monosaccharides & oligosaccharide are crystalline solids, soluble in water & sweet in taste. These are collectively known as Sugars.
- Polysaccharides are amorphous, sparingly soluble in water & tasteless & are known as Non-Sugars.

Reducing & Non Reducing Sugars :-

- The carbohydrates that reduce Fehling's reagent, Tollen's reagent, are called as reducing sugars.
- All monosaccharides whether aldose or ketose, are reducing sugars. Eg :- Glucose, Fructose, mannose, galactose.

Monosaccharides :-

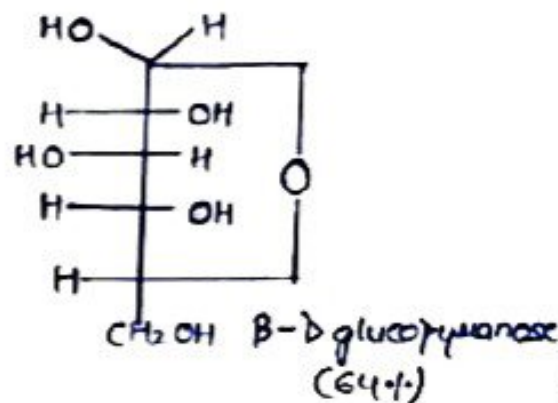
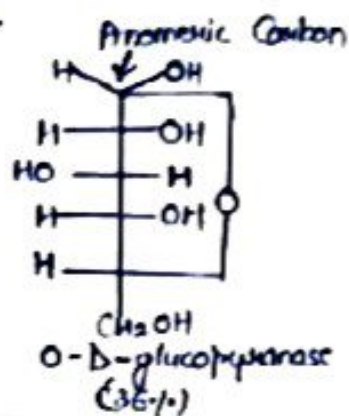
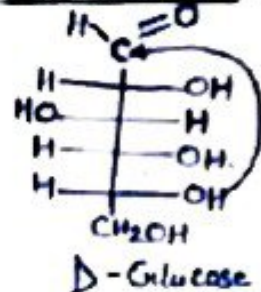
(1) Glucose ($C_6H_{12}O_6$) :-

- It is act as a reducing agent (reduces both Fehling's solution & ammoniacal silver nitrate solution).

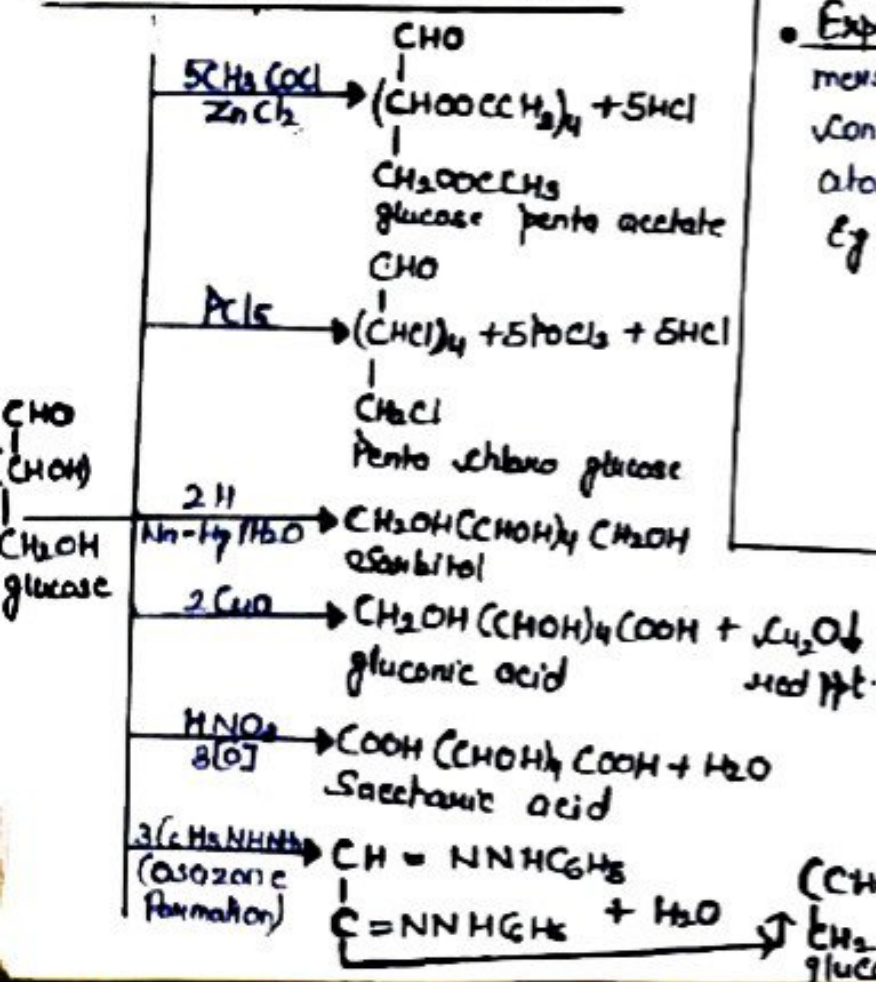
Physical properties :-

- Glucose is sweet in taste & also optically active (dextro rotatory).
- Glucose shows mutarotation.

Structure :-

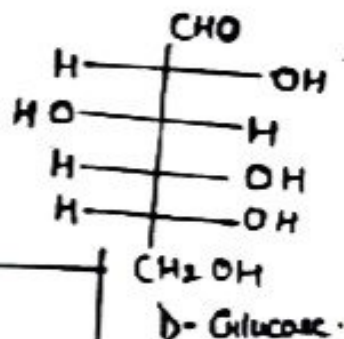


Chemical Reaction of Glucose :-



- Epimers :- are a pair of diastereomers that differ only in the configuration about a single carbon atom.

Eg :- Glucose & Mannose are C2 epimers



Reducing & Non Reducing Sugars :-

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Monosaccharides :-

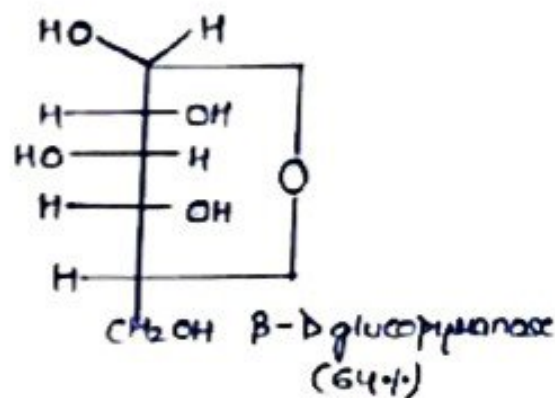
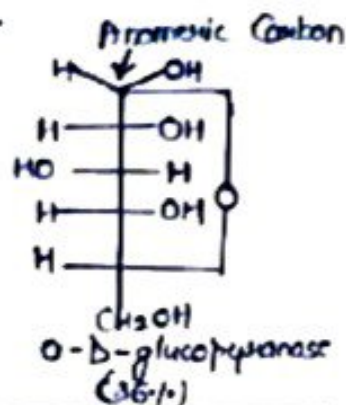
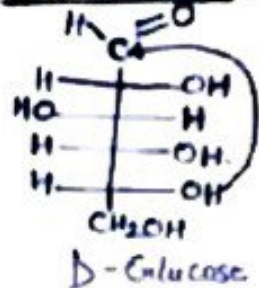
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- It acts as a reducing agent (reduces both Fehling's solution & ammoniacal silver nitrate solution).

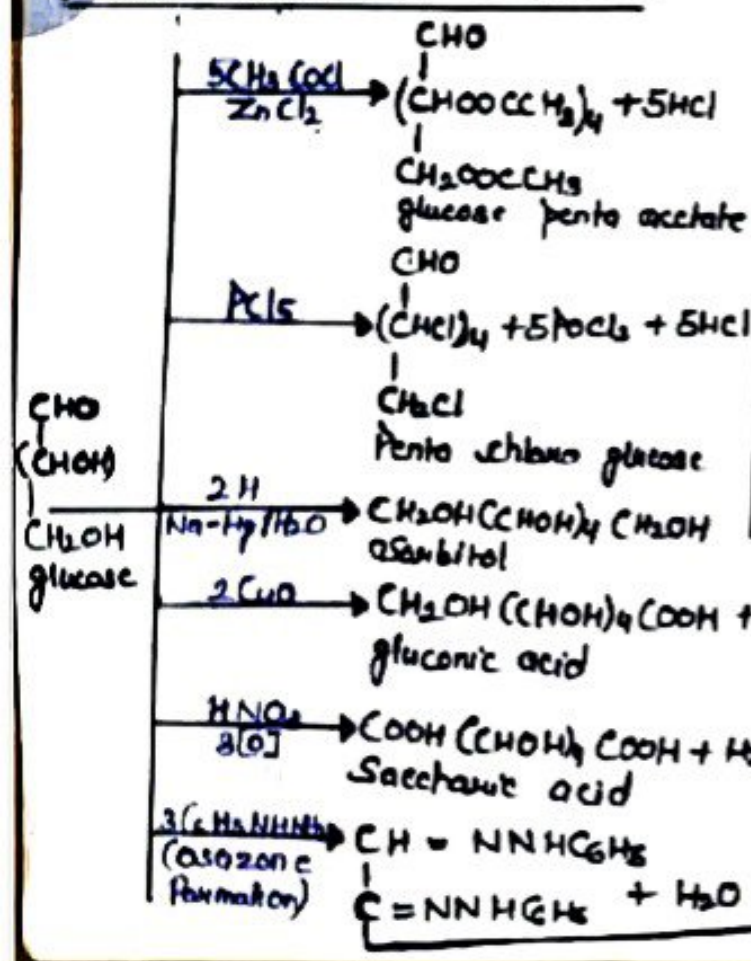
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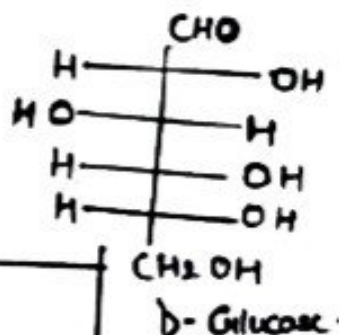


Chemical Reaction of Glucose :-



- Epimers :- are a pair of diastereoisomers that differ only in the configuration about a single carbon atom.

Eg :- Glucose & Mannose are C₂ epimers



Cyclic Structure of Glucose :-

- The open chain structure of glucose proposed by Baeyer explained most of its properties. But it could not explain the following:-
 - 1) Glucose does not give Schiff's test & does not react with NaHSO_3 & give hydrogensulphite addition product inspite of presence of $-\text{CHO}$ group.
 - 2) Pentacetate of glucose does not react with NH_2OH group indicating absence of $-\text{CHO}$ group.

Mutautation of glucose :-

- This spontaneous change in specific rotations of an optically active compound is called mutautation.
- These properties can be explained by cyclic structure of glucose. Glucose forms a stable cyclic hemiacetal. Initially five membered ring structure of glucose is proposed & it known as Pyranose.

Anomers :-

Anomers are diastereomers that differ in the configuration at the acetal or hemiacetal C atom of a sugar in its cyclic form. or Anomers are epimers whose conformations differ only about C-1.

For eg $\alpha\text{-D}(+) \text{ \& } \beta\text{-D}(+) \text{ glucose}$ are anomers.

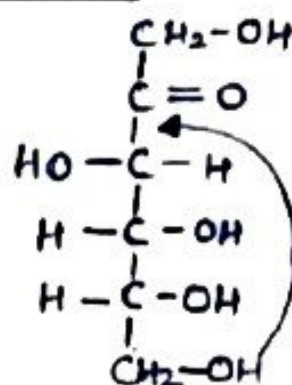
$\alpha\text{-D}(-) \text{ \& } \beta\text{-D}(-) \text{ fructose}$ are anomers.

- In glucose C1 carbon is anomeric carbon & in fructose C2 carbon is anomeric carbon.

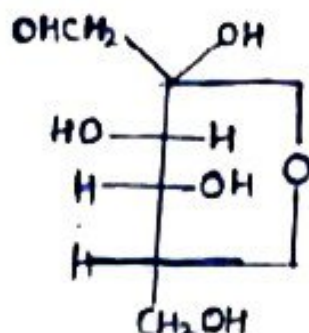
(2) Fructose :-

- It is also known as $\alpha\text{-Levulose}$ i.e. natural occurring compound is laevorotatory.
- It is pentahydroxy ketone & shows mutautation like glucose.
- It is reducing sugars.

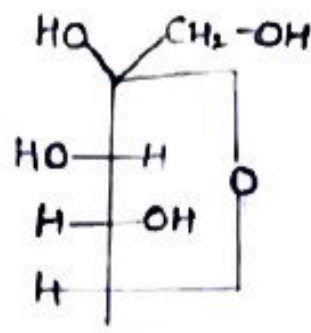
Structure :-



Open chain structure
of fructose
Specific Rotation (-92°)



$\alpha\text{-D-fructofuranose}$
Specific Rotation
 (-21°)



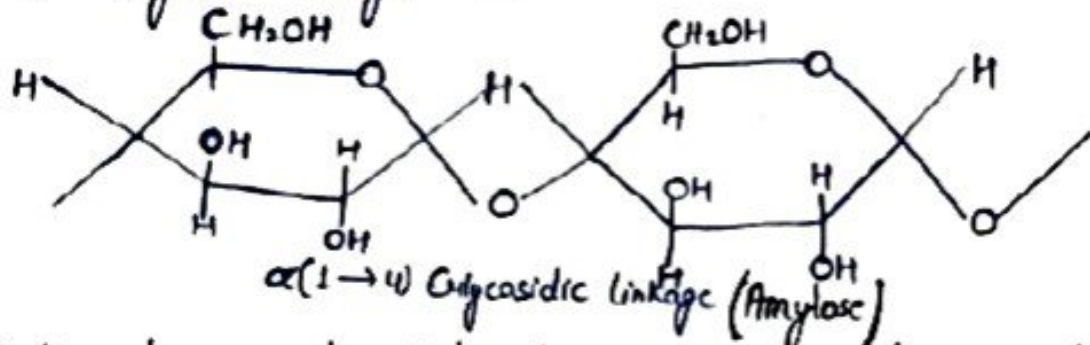
$\beta\text{-D-fructofuranose}$
Specific Rotation
 (-133°)

Polysaccharides :-

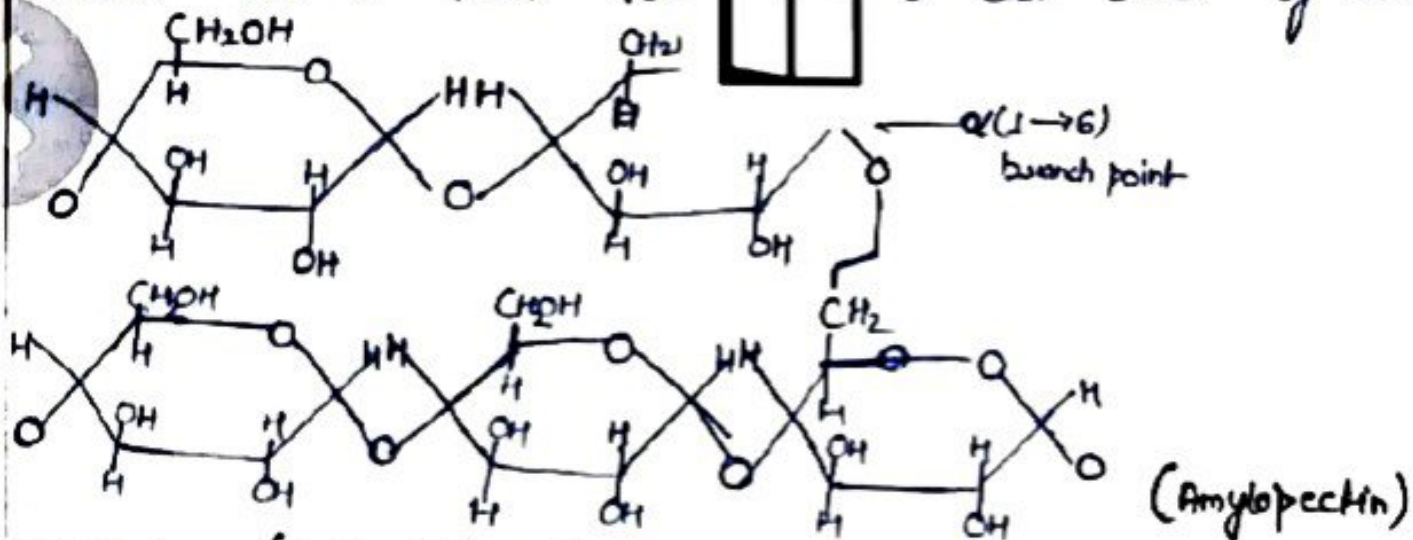
(A) Starch ($C_6H_{10}O_5$)_n :-

Starch $\rightarrow$ Amylose (20%) (water soluble)
 $\rightarrow$ Amylopectin (80%) (water insoluble).

- Both amylose & amylopectin are composed of D-glucose units.
- The amylose molecule is made up of D-glucose unit joined by α -glycosidic linkages between C-1 of one glucose unit & C-4 of the next glucose unit. The no. of D-glucose units in amylose range from 60-300.

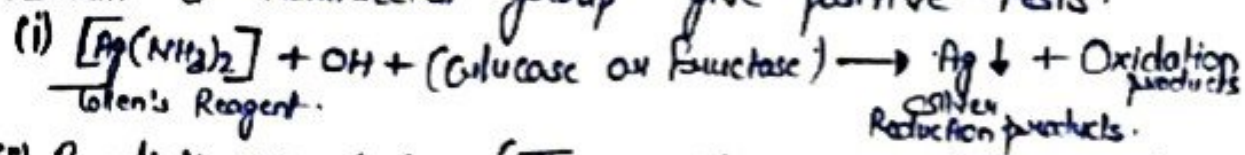


- Amylopectin has a branched-chain structure, linkages between C-1 to one glucose unit & C-4 next glucose unit. These chains are in turn α to each other by 1,6-linkages.

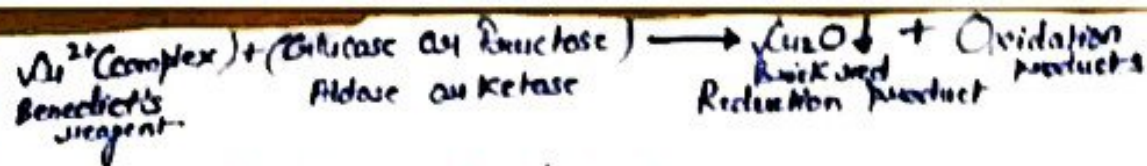


(B) Cellulose ($C_6H_{10}O_5$)_n :-

- It is non-reducing sugar.
- It is a straight chain of β -D-glycosidic linkage. $\rightarrow$ Tollen's, Fehling's & Benedict's Reagents.
- Sugars which give positive tests with these reagents are known as reducing sugars & all the carbohydrates which contain a hemiacetal group give positive tests.



(ii) Benedict's reagent :- (They oxidise an aldose or ketose & give brick red precipitates of Cu_2O .)



Amino Acids & Proteins :-

Amino Acid :-

- The bond between two amino acid molecules is peptide bond 'or' amide bond & the resultant is known as dipeptide.
 - All protein are polymers of α -amino acids & on partial hydrolysis give peptides of varying molecular masses which upon complete hydrolysis give α -amino acids.
- proteins $\xrightarrow{\text{hydrolysis}}$ peptides $\xrightarrow{\text{hydrolysis}}$ α -amino acids.

Classification of Amino acid :-

Essential Amino acid :- Amino acids which are necessary be present in our diet plan.

Non-Essential Amino acid :- 10 amino acids that are synthesis in our body & these are avoid to be non-essential acids.

(i) Essential Amino acid :-

Eg (1) Leucine (2) Isoleucine (3) Lysine.

(ii) Non-Essential Amino acid :-

(1) Alanine (2) Asparagine (3) Aspartic acid (4) Cysteine.

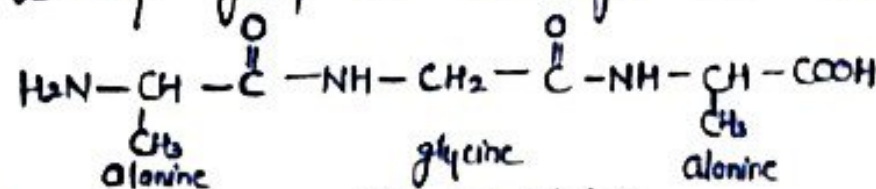
(iii) Properties :-

Except glycine all other naturally occurring α -amino acids are optically active, since α -carbon is asymmetric. L-amino acids are represented by writing the $-\text{NH}_2$ group on the left hand side.

Acidic linkage :- Amino acid units are linked together by peptide

$-\text{C}-\text{NH}-$ linkage & form polypeptides.

- In a polypeptide structure free amino group (NH_2) i.e. N-terminal residue is written on the left hand side & the free carboxyl group on the right hand side of the chain.



Alanyl glycyl alanine.

In the above structure $-\text{COOH}$ group is C-terminal residue & group is N-terminal residue & group is N-terminal residue.

Proteins :-

- These are high molecular mass complex, biopolymers of amino acids

Classification of proteins :-

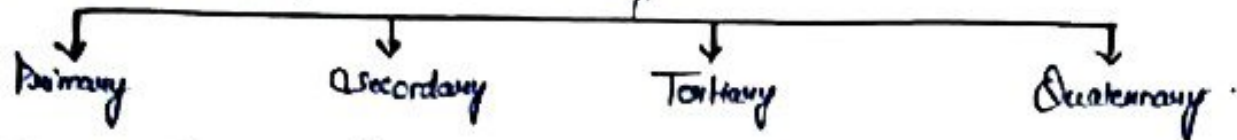
(i) On the basis of molecular structure :-

Fibrous proteins :- Eg Keratin, myosin, collagen etc.

Globular proteins :- Eg Insulin, albumin, thyroglobulin, antibodies, haemoglobin, fibrinogen etc.

Structure of proteins :-

This conformation is of 4 types :-



1. Primary structure :-

- Primary structure is conformed by a single polypeptide chain in a linear manner.

2. Secondary structure :-

- The conformation which the polypeptide chain assume as a result of H-bonding is called secondary structure of protein.
- The H-bonds are present between hydrogen of amino group & oxygen atom of carboxylic of protein.
- The structure is of two types :-

α -Helix

β -pleated sheet

α -Helix :- Intermolecular H-bonds are present Eg. Myosin, Keratin etc.

β -pleated sheet :- Intermolecular H-bonds hold together the neighbouring polypeptide chains. Eg. Silk fibres.

3. Tertiary structure :-

- Tertiary structure refers to its three dimensional structure i.e. folding & bonding of the long peptide chains.
 - This structure is formed by 4 types of bonds.
- (i) Hydrogen bond : (ii) Hydrophobic bond : (iii) Ionic bond : (iv) Disulphide bond.

(4) Quaternary structure :-

When two or more polypeptide chains unite by forces other than covalent bonds (i.e. not peptide & disulphide bonds) we get quaternary structure of protein. It is most stable structure.

Eg :- Haemoglobin.

- Changing the pH denatures proteins because it changes the charges on many of the side chains. This disrupts electrostatic attractions & hydrogen bonds.
- The coagulation of egg white on boiling & curdling of milk caused by the bacteria present in milk are common eg. of denaturation of protein.

Hormones :-

These are water soluble hormones containing groups. These are derived from amino hydroxide & adrenaline.

Eg :-

(1) Epinephrine or Adrenaline is a hormone that helps to control blood pressure & increase pulse rate.

It helps to reduce fatty acids from fat & glucose from liver glycogen.

(2) Thyroxine hormones secreted by Thyroid.

It regulates metabolism of lipids, proteins & carbohydrates.

(3) Testosterone Regulates & stimulates male sex organs.

Vitamins :-

Vitamins may be defined as a group of biomolecules which are required in small amounts for normal metabolic process & for the life, growth & health of human beings & animals.

Classification of Vitamins :-

Vitamins are broadly classified into two categories :-

(1) Water Soluble Vitamins :- Vitamin B-complex & Vitamin C, are water soluble vitamins & must be supplied regularly in diet.

(2) Fat soluble vitamins :- These are oily substances & soluble in fat. These are A, D, E & K. They are stored in liver & adipose (fat storing) tissues.

(3) Biotin (Vitamin H) :- It is neither soluble in water nor in fats. Lack of particular vitamin causes a specific deficiency disease.

Eg :- of some important vitamins & deficiency disease are :-

Vitamin A (Retinol) :- Xerophthalmia (hardening of eye). causes night blindness & xerosis (drying of skin).

Vitamin B₁ (Thiamine) :- Beriberi (paralysis of legs & general weakness) loss of appetite.

Vitamin D (Ergocalciferol) (Sunshine Vitamin) :- Rickets (deformation of bones) Osteomalacia (soft bones & joint pain).

Vitamin C :- Is chemically known as ascorbic acid - Scurvy.

Nucleic Acid :-

Nucleic acids are colourless, complex, amorphous compounds

made up of three units: bases, sugar & phosphoric acid. These are obtained by the hydrolysis of nucleoproteins which is a class of conjugated.

Nucleic acids are of two types:-

- (i) Pentose nucleic acids or ribonucleic acids (R.N.A).
- (ii) Deoxypentose nucleic acids or deoxyribonucleic acids (D.N.A).

• Nucleic acids can be hydrolysed in stages to nucleotides, nucleosides & phosphoric acid & ultimately to base & sugar.

Nucleic acid $\rightarrow$ Nucleotides $\rightarrow$ Nucleosides + H_3PO_4

$\downarrow$
Base + Sugar

Base:- Important purine bases are adenine & guanine; while pyrimidine bases are uracil, thymine & cytosine.

• Adenine, guanine & cytosine are present in RNA as well as in DNA, while thymine is present only in DNA & uracil only in RNA.

Nucleoside:-

- Each nucleoside consists of sugar molecule & a nitrogenous base.

The relationship can be shown as given below.

Nucleic acid = many nucleotides

Nucleotide = nucleosides + phosphate

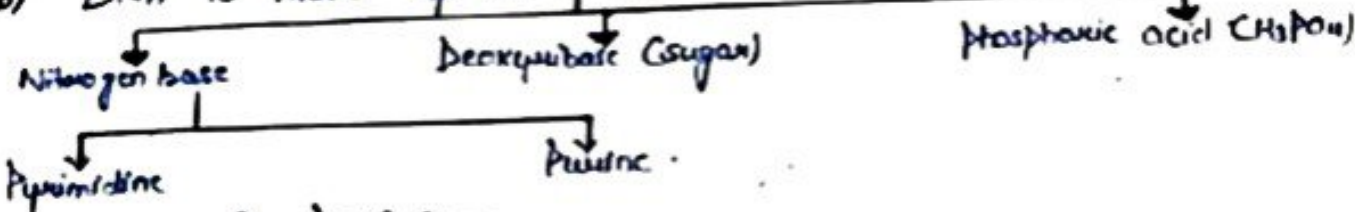
Nucleoside = Sugar + nitrogenous base.

Thus nucleotide = phosphate + sugar + hydrogenous base.

Deoxyribose Nucleic Acid (DNA):-

(a) It is found in nucleus.

(b) DNA is made up of 3 units -



Structure of DNA:-

• DNA has a double helix structure & is made up of two chains of polynucleotides.

• DNA is a polymer of deoxynucleotides.

• The two strands are joined by 3' $\rightarrow$ 5' phosphodiester bonds.

• Sugar & phosphates are alternately arranged.

• In both chains in between A & T, 2 hydrogen bonds ($A \equiv \equiv T$) are present, while in C & G, 3 H-bonds ($C \equiv \equiv G$) are present.

• A always attaches with T while C always attaches with G.

Functions of DNA :-

- (i) Self-replication or self-duplication
- (ii) Protein synthesis.

The specific sequence of base pairs in DNA represents coded information for the manufacture of specific proteins. T.

Ribonucleic Acid (RNA) :-

Ribonucleic acid is a polymer of purine & pyrimidine ribonucleotides linked by $3' \rightarrow 5'$ phosphodiester bridges. RNA exists basically as a single-stranded molecule rather than as a double-stranded helical molecule, as does DNA, H₂RNA (m-RNA).